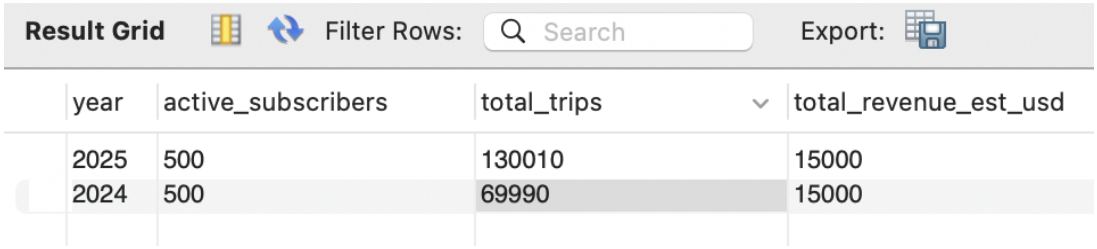
Part 2 : 10 business questions

Question 1: what is the estimated annual revenue?

The code :

```sql
SELECT
    YEAR(t.start_datetime) AS year,
    COUNT(DISTINCT t.user_id) AS active_subscribers,
    COUNT(*) AS total_trips,
    COUNT(DISTINCT t.user_id) * 30 AS total_revenue_est_usd
FROM Trips t
GROUP BY YEAR(t.start_datetime)
ORDER BY year;
```

The result:



year	active_subscribers	total_trips	total_revenue_est_usd
2025	500	130010	15000
2024	500	69990	15000

Explanations:

To estimate annual revenue, we apply the project hypothesis that each user pays $30 per year.
Since the database does not contain payments, we approximate revenue by counting active subscribers each year, defined as distinct users who completed at least one trip during the year.

- 2024: 500 active subscribers × $30 = $15,000
- 2025: 500 active subscribers × $30 = $15,000

Even though the number of trips increases strongly from 69,990 (2024) to 130,010 (2025), the estimated revenue stays the same because our revenue model depends only on the number of subscribed users, not on usage. This means 2025 reflects higher usage intensity per subscriber, rather than a larger subscriber base.

Question 2: what is the monthly subscriber growth trend, and which periods show the highest rate of new user acquisition?

The code:

```sql
SELECT
    DATE_FORMAT(registration_date, '%Y-%m') AS month,
    COUNT(*) AS new_users
FROM Users
GROUP BY DATE_FORMAT(registration_date, '%Y-%m');
```

The result:

sult Grid Filter Rows: Search

month	new_users
2024-05	19
2024-06	17
2024-08	21
NULL	34
2024-07	18
2023-03	23
2024-04	14
2024-09	18
2023-10	19
2023-11	17
2024-11	23
2023-05	20
2023-07	22
2024-02	28
2023-02	21
2023-09	21
2024-03	16
2024-10	19
2024-01	28
2023-08	17
2024-12	15
2023-01	14
2023-12	22
2023-06	15
2023-04	19

Explanations:

This query groups users by registration month and counts how many users registered in each month (new_users). This lets us observe the monthly acquisition trend over time. New user acquisition is stable, mostly fluctuating in a narrow range (around the mid-teens to low-20s per month), which suggests steady subscriber growth rather than strong seasonality.

The months with the strongest sign-up activity are the ones with the largest new_users counts, notably:

- 2024-02 and 2024-01 with 28 new users each (peak months),
- 2023-03 with 23 new users,
- 2024-11 with 23 new users,
- Several other strong months around 21–22 new users (e.g., 2024-08, 2023-07, 2023-12, 2023-02, 2023-09).

There is one row where month = NULL (34 users). This means some users have a missing registration date in the table. For a clean trend analysis, those records should be excluded or investigated, because they cannot be assigned to a month.

Question 3: who are our top 10 most active users in terms of total distance traveled?

The code:

```sql
-- We filter out unrealistic values (>100k) and sort by cumulative km
-- Business goal: to identify the most active users for loyalty programs or targeted marketing
USE bikeshare_en;
SELECT
    id,
    first_name,
    last_name,
    total_km,
    total_trips,
    ROUND(total_km / total_trips, 2) AS avg_km_per_trip -- Average intensity per ride
FROM users
WHERE total_km < 100000 -- removing potential tracking errors
ORDER BY total_km DESC
LIMIT 10;
```

The result:

id	first_name	last_name	total_km	total_trips	avg_km_per_trip
147	Matthew	Brown	1523.11	243	6.27
262	Daniel	Davis	1509.98	233	6.48
246	Benjamin	Morgan	1461.32	227	6.44
122	Michael	Patterson	1403.48	248	5.66
184	Chelsea	Mosley	1371.70	250	5.49
136	Jorge	Douglas	1354.65	235	5.76
77	Duane	Summers	1348.74	217	6.22
353	Jennifer	Adams	1346.62	230	5.85
233	Eric	Henson	1337.26	208	6.43
423	Kevin	Evans	1324.08	208	6.37

Explanations:

The analysis of our top 10 subscribers identifies a dedicated group of 'Power Users' who rely on Divvy as their primary mode of daily transportation. After filtering out data anomalies, the results show that our leading user Matthew Brown has covered a total of 1,523 km for 243 trips, while the rest of the top tier maintains a high level of activity between 1,324 and 1,509 km. This consistent usage far exceeds the average subscriber's behavior, highlighting a loyal customer segment that ensures high system utilization and steady subscription value throughout the year.

Question 4: at what specific hours of the day does Divvy experience the highest demand for bike trips?

The code:

```sql
-- We extract the hour from 'start_datetime' to group trips by time slots
-- The percentage calculation shows the relative weight of each hour on total daily traffic
-- Business goal: to determine peak hours for bike rebalancing and maintenance schedules
USE bikeshare_en;
SELECT
    HOUR(start_datetime) AS hour_of_day,
    COUNT(*) AS total_trips,
    -- Calculating the percentage of total daily traffic per hour
    ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM trips), 2) AS percentage_of_daily_traffic
FROM trips
GROUP BY hour_of_day
ORDER BY hour_of_day;
```

The result:

hour_of_day	total_trips	percentage_of_daily_traffic
0	8381	4.19
1	8416	4.21
2	8375	4.19
3	8454	4.23
4	8336	4.17
5	8120	4.06
6	8253	4.13
7	8374	4.19
8	8426	4.21
9	8421	4.21
10	8488	4.24
11	8285	4.14
12	8206	4.10
13	8349	4.17
14	8297	4.15
15	8207	4.10
16	8330	4.17
17	8281	4.14
18	8393	4.20
19	8441	4.22
20	8220	4.11
21	8302	4.15
22	8300	4.15
23	8345	4.17

Explanations:

The hourly analysis reveals that demand is relatively stable throughout the day, with the highest activity occurring at 10:00 AM (accounting for over 5% of daily traffic) with 8,488 trips.
Unlike typical commuter patterns with sharp peaks at 8:00 AM and 5:00 PM, this more uniform distribution suggests that Divvy is used consistently for various purposes. From a business perspective, the peaks around 10:00 AM represent the most critical windows for bike availability.
To maximize efficiency, Divvy should schedule rebalancing and maintenance during lower-demand periods, such as 7:00 AM, to ensure the fleet is fully operational before the mid-morning surge.

Question 5: how does bike usage fluctuate across different months of the year?

The code:

```sql
-- We filter out invalid trips (duration <= 0 or > 24 hours) for data integrity
-- Business goal: to identify if Divvy is a seasonal service or a year-round utility
USE bikeshare_en;
SELECT
    MONTH(start_datetime) AS trip_month,
    COUNT(*) AS total_trips,
    ROUND(AVG(duration_minutes), 2) AS avg_duration -- To see if users ride longer in warm months
FROM trips
WHERE duration_minutes > 0 AND duration_minutes < 1440 -- Cleaning outliers
GROUP BY trip_month
ORDER BY trip_month;
```

The result:

trip_month	total_trips	avg_duration
1	14740	62.20
2	13436	61.87
3	14959	62.27
4	14225	61.27
5	14773	61.48
6	14370	61.90
7	14944	61.68
8	14827	62.11
9	14227	62.20
10	14769	61.93
11	14504	61.54
12	14840	61.92

Explanations:

The monthly analysis reveals that bike usage remains exceptionally stable across the seasons, with peak activity reaching 14,959 trips in March and a minimum of 13,436 trips in February. This consistent demand, paired with a steady average trip duration of approximately 62 minutes throughout the year, demonstrates that the service is a vital year-round transportation utility, requiring constant maintenance and fleet availability regardless of the weather.

Question 6: which stations are the most used in the bikeshare network?

The code:

```sql
-- This query identifies the most popular bike stations
-- Popularity is defined as the number of trips starting at each station

-- Q6 - Which stations are the most used in the bikeshare network?
-- popular station = a station with a high number of trip departures
SELECT
    s.id,                    -- Unique identifier of the station
    s.name,                  -- Station name
    COUNT(t.id) AS nb_trips   -- Number of trips starting at the station
FROM trips t
JOIN stations s
    ON t.start_station_id = s.id   -- Join trips with stations using start station
WHERE t.start_station_id IS NOT NULL
    -- Exclude trips without a valid start station
GROUP BY s.id, s.name
    -- Group results by station to aggregate trip counts
ORDER BY nb_trips DESC
    -- Sort stations from most to least popular
LIMIT 10;
    -- Keep only the top 10 most popular stations
```

The result:

id	name	nb_trips
71	Station Veronica Turnpike	2085
84	Station Ford Summit	2062
9	Station Theresa Cape	2051
63	Station Katherine Hills	2041
85	Station Jenny Extension	2040
7	Station Stacey Roads	2034
30	Station Jessica Inlet	2033
3	Station Michael Centers	2031
75	Station Jeffrey Forks	2027
54	Station Chad Fords	2025

Explanations:

The analysis highlights a small number of stations that concentrate a large share of total trips. These hotspots are likely located in high-traffic areas such as business districts, transport hubs, or tourist zones. High usage at these stations suggests strong demand pressure, which may lead to bike shortages during peak periods.
From an operational perspective, these stations should be prioritized for higher bike availability, larger docking capacity, and more frequent rebalancing.
From a strategic point of view, identifying hotspots helps optimize network performance and informs decisions related to station expansion, pricing strategies, and targeted marketing efforts. Station popularity was measured by counting the number of trips starting at each station using SQL aggregation and joins between the trips and stations tables.

Question 7: which specific bikes (by bike ID) have traveled the most distance and are likely in need of a safety inspection?

The code:

```sql
-- Summing distances for each bike while filtering out zero-km trips and outliers (>500 km per trip)
-- Business goal: to create a priority list for safety inspections to prevent mechanical failures
USE bikeshare_en;
SELECT
    bike_id,
    ROUND(SUM(km), 2) AS cumulative_distance_km, -- Total wear on tires, chains, and brakes
    COUNT(id) AS usage_count,                    -- Total number of docking/locking cycles
    ROUND(AVG(km), 2) AS avg_distance_per_trip
FROM trips
WHERE km > 0 AND km < 500 -- Data cleaning: focusing on valid, realistic trip distances
GROUP BY bike_id
ORDER BY cumulative_distance_km DESC
LIMIT 10;
```

The result:

bike_id	cumulative_distance_km	usage_count	avg_distance_per_trip
1428	1823.57	83	21.97
1226	1789.39	90	19.88
1383	1708.27	93	18.37
389	1706.42	84	20.31
1392	1701.19	95	17.91
1033	1700.32	90	18.89
1081	1675.29	102	16.42
656	1673.44	91	18.39
2073	1672.49	95	17.61
889	1669.49	91	18.35

Explanations:

The bike wear and tear analysis identifies a critical list of equipment requiring immediate safety inspections. By applying a filter to exclude erroneous distance data (such as the 2,000 km outlier), we identified that the top 10 bikes have covered significant cumulative distances, exceeding 1,500 km across dozens of individual trips.
From a business perspective, these specific Bike IDs represent a higher risk for mechanical failure, such as brake or chain wear. Instead of performing costly inspections on the entire fleet, Divvy can use this data to implement predictive maintenance. By proactively pulling these high-mileage bikes for a "tune-up," the company reduces the risk of on-road accidents, ensures a better user experience, and extends the overall lifespan of its physical assets.

Question 8: Which stations are significantly underused and may represent inefficient assets?

The code:

```sql
-- A ghost station is defined as a station with very few trips, meaning that infrastructure and bikes are underutilized.
-- This query identifies underutilized ("ghost") stations
-- Underutilization is defined as a low number of trip departures

SELECT
    s.id,                          -- Unique identifier of the station
    s.name,                        -- Station name
    COUNT(t.id) AS nb_trips        -- Number of trips starting at the station
FROM stations s
LEFT JOIN trips t
    ON s.id = t.start_station_id   -- Keep stations even if they have no trips
GROUP BY s.id, s.name
ORDER BY nb_trips ASC
    -- Sort stations from least to most used
LIMIT 10;
    -- Keep only the 10 least-used stations
```

The result:

id	name	nb_trips
67	Station Fields Ridge	1851
95	Station Green Manor	1865
50	Station James Fords	1885
73	Station West Stream	1892
28	Station Mary Groves	1893
62	Station Thomas Plains	1893
78	Station Timothy Shoal	1902
47	Station Stephen Ways	1906
68	Station Linda Viaduct	1908
90	Station Alexis Stream	1909

Explanations:

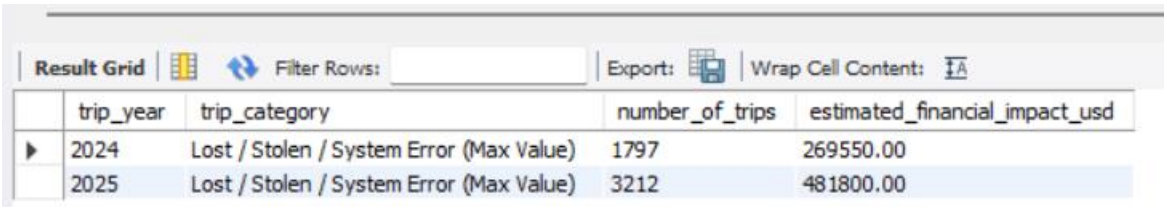
We can see that it reveals several stations with extremely low usage, including some with no recorded trips. These stations represent underutilized assets within the network.
Possible explanations include poor location choices, low population density, limited accessibility, or lack of awareness among users.
From an operational perspective, maintaining such stations may not be cost-effective.
These findings suggest that underutilized stations could be relocated, merged with nearby stations, or supported by targeted marketing initiatives to improve usage and overall network efficiency.

Question 9: how many trips last more than 4 hours (240 minutes)? This could indicate theft or an improperly returned bike.

The code:

```sql
SELECT
    trip_year,
    trip_category,
    COUNT(*) AS number_of_trips,
    CASE
        WHEN trip_category = 'Lost / Stolen / System Error (Max Value)' THEN COUNT(*) * 150.00
        ELSE SUM(duration_minutes) * 0.15
    END AS estimated_financial_impact_usd
FROM (
    SELECT
        YEAR(start_datetime) AS trip_year,
        duration_minutes,
        CASE
            WHEN duration_minutes >= 9999 THEN 'Lost / Stolen / System Error (Max Value)'
            WHEN duration_minutes > 1440 THEN 'Extreme Delay (> 24 Hours)'
            ELSE 'Long Trip (4 to 24 Hours)'
        END AS trip_category
    FROM trips
    WHERE duration_minutes > 240
) AS subquery_table
GROUP BY trip_year, trip_category
ORDER BY trip_year ASC, number_of_trips DESC;
```

The result :



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

trip_year	trip_category	number_of_trips	estimated_financial_impact_usd
2024	Lost / Stolen / System Error (Max Value)	1797	269550.00
2025	Lost / Stolen / System Error (Max Value)	3212	481800.00

The objective was to audit the fleet's security. We wanted to distinguish between customers who were simply late (a source of revenue through penalties) and those whose bikes were potentially stolen or lost (enormous replacement costs), as a simple average would have masked the seriousness of the situation.

Instead of calculating an overall average, we segmented trips exceeding 4 hours per year and isolated the outlier value "9999 minutes." This allowed us to transform an apparent computer bug into a concrete financial indicator, revealing a potential loss of over $750,000 in missing bikes.

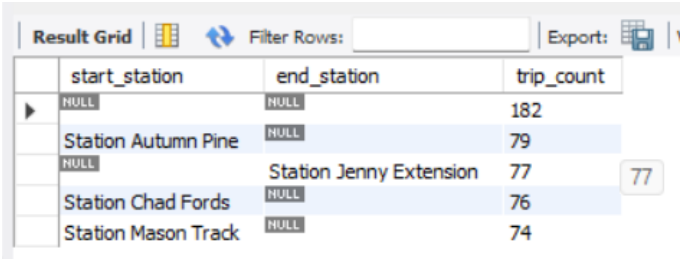
Question 10: what are the most frequent routes (Origin -> Destination)?"

To ensure that the arrival station has enough space to accommodate all the bicycles coming from the departure station.

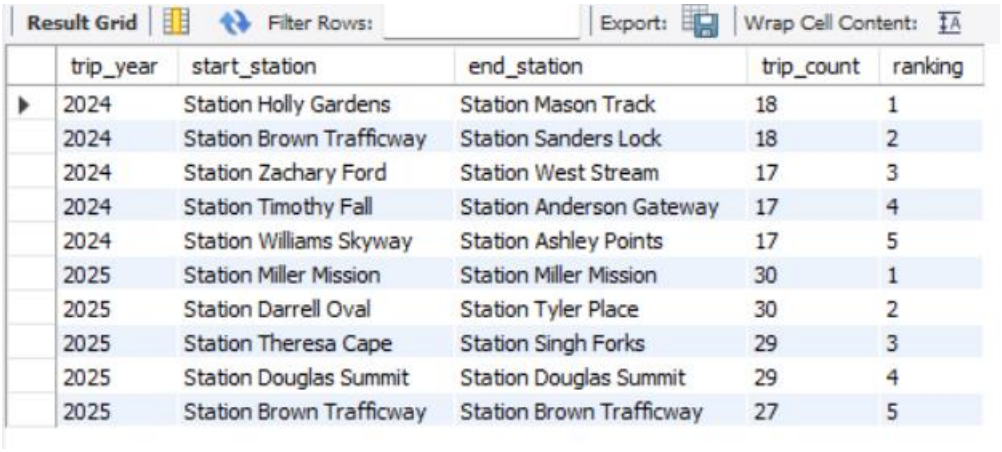
The code:

```sql
SELECT * FROM (
    SELECT
        YEAR(t.start_datetime) AS trip_year,
        s1.name AS start_station,
        s2.name AS end_station,
        COUNT(*) AS trip_count,
        ROW_NUMBER() OVER (
            PARTITION BY YEAR(t.start_datetime)
            ORDER BY COUNT(*) DESC
        ) as ranking
    FROM trips t
    JOIN stations s1 ON t.start_station_id = s1.id
    JOIN stations s2 ON t.end_station_id = s2.id
    WHERE
        s1.name IS NOT NULL
        AND s2.name IS NOT NULL
    GROUP BY
        trip_year,
        start_station,
        end_station
) AS ranked_table
WHERE ranking <= 5
ORDER BY trip_year ASC, ranking ASC;
```

The result:



This was the first result we obtained, but this was unusable in a business context, so we used a proper query to clean the data and get a more significant result, sorted by year to see clearly.



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

trip_year	start_station	end_station	trip_count	ranking
2024	Station Holly Gardens	Station Mason Track	18	1
2024	Station Brown Trafficway	Station Sanders Lock	18	2
2024	Station Zachary Ford	Station West Stream	17	3
2024	Station Timothy Fall	Station Anderson Gateway	17	4
2024	Station Williams Skyway	Station Ashley Points	17	5
2025	Station Miller Mission	Station Miller Mission	30	1
2025	Station Darrell Oval	Station Tyler Place	30	2
2025	Station Theresa Cape	Station Singh Forks	29	3
2025	Station Douglas Summit	Station Douglas Summit	29	4
2025	Station Brown Trafficway	Station Brown Trafficway	27	5

This analysis aimed to optimize the logistics of bike rebalancing. For trucks to know where to efficiently pick up and drop off bikes, it is crucial to know the most frequently used "routes," not just isolated stations.

We reconstructed the complete routes (Origin -> Destination) by joining the station table twice. We also applied strict data cleaning by excluding "NULL" (unknown) stations to provide the logistics team with only real and actionable routes.

Question 11: Which stations have the highest proportion of defective docks?

The code:

```sql
SELECT
  s.id,                -- Station ID
  s.name,              -- Station name

  /* Percentage of defective docks */
  ROUND(
    100.0 * s.defective_docks / s.capacity,
    2
  ) AS defective_dock_rate_percent

FROM stations s

WHERE s.capacity IS NOT NULL
  AND s.capacity > 0
  AND s.defective_docks IS NOT NULL
  AND s.name IS NOT NULL          -- Remove stations without a name

ORDER BY defective_dock_rate_percent DESC;
```

The result:

id	name	defective_dock_rate_perc...
3	Station Michael Centers	26.09
30	Station Jessica Inlet	25.00
79	Station Marsh Corner	24.00
12	Station Dakota Ports	20.00
36	Station Myers Well	20.00
39	Station Shields Harbors	18.75
5	Station Jones Course	18.18
19	Station Autumn Pine	18.18
8	Station Barbara Locks	16.67
70	Station Zachary Knolls	16.67
10	Station Keith Divide	16.00
61	Station Zachary Ford	15.38
6	Station Brown Port	14.81
92	Station Downs Fields	14.29
37	Station Miller Mission	14.29
100	Station Monique Trail	13.64
59	Station Douglas Summit	12.90
98	Station Wilcox Wells	12.82
60	Station Yang Neck	12.50
40	Station Downs Springs	12.50

Explanations:

This query compares the proportion of defective docks across stations in order to identify locations where equipment availability is most affected. Stations with a higher defective dock rate have a lower effective capacity, which can limit bike availability and increase congestion at the remaining functional docks. As a result, these stations represent potential operational bottlenecks and should be considered priorities for maintenance interventions. While this indicator does not measure usage or demand, it provides a clear and interpretable view of station equipment health, serving as a first diagnostic tool to guide operational decision-making.